WORKED SOLUTION

STEP I 2018, Question 2

Pure • Logarithmic Inequalities

STEP I 2018, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q2

The key idea

Everything rests on the change-of-base rule: a logarithm in any base is a ratio of logarithms in a common base, and $\frac{1}{\log_b c} = \log_c b$. Once every logarithm is rewritten in terms of ln, each part becomes a short piece of algebra with inequalities. The given facts such as $e^2 < 8$ are just disguised bounds on $\ln 2$ and $\ln 10$. All the while we use the fact that ln is increasing, so taking logarithms keeps the direction of an inequality between positive numbers.

Solution

The opening result

By the definition of a logarithm, $x = \log_b c$ means

$$c = b^x.$$

Taking logarithms to base a of both sides and using $\log_a(b^x) = x \log_a b$,

$$\log_a c = x \log_a b.$$

Here $b \neq 1$ (so that $\log_b$ makes sense), so $\log_a b \neq 0$ and we may divide:

$$\frac{\log_a c}{\log_a b} = x = \log_b c,$$

as required. Two consequences are used below. Taking $a = c$ gives $\log_b c = \dfrac{1}{\log_c b}$, and taking $a = e$ gives $\log_b c = \dfrac{\ln c}{\ln b}$.

Part (i)

By the reciprocal rule, $\dfrac{1}{\log_2 \pi} = \log_\pi 2$ and $\dfrac{1}{\log_5 \pi} = \log_\pi 5$. So

$$\frac{1}{\log_2 \pi} + \frac{1}{\log_5 \pi} = \log_\pi 2 + \log_\pi 5 = \log_\pi 10.$$

Since $\pi > 1$, the function $\log_\pi$ is increasing, and $10 > \pi^2$ gives

$$\log_\pi 10 > \log_\pi \pi^2 = 2,$$

which is the required inequality.

Part (ii)

Write the first given fact in natural logarithms:

$$\log_2 \frac{\pi}{e} = \frac{\ln \pi - 1}{\ln 2} > \frac{1}{5} \quad \Longrightarrow \quad \ln \pi > 1 + \frac{\ln 2}{5},$$

where we multiplied by $\ln 2 > 0$, so the inequality keeps its direction.

Ask yourself What is $e^2 < 8$ telling us?

It is a lower bound for $\ln 2$. Taking ln of both sides (allowed, as ln is increasing) gives $2 < \ln 8 = 3 \ln 2$, so $\ln 2 > \frac{2}{3}$. Substituting,

$$\ln \pi > 1 + \frac{1}{5} \cdot \frac{2}{3} = 1 + \frac{2}{15} = \frac{17}{15},$$

as required.

Part (iii)

From $\pi^2 < 10$, taking logarithms gives $2 \ln \pi < \ln 10$. So it is enough to show $\ln 10 < \frac{30}{13}$. Translate the other two facts. Since $e^3 > 20$, we have $3 > \ln 20 = \ln 2 + \ln 10$, that is,

$$\ln 2 < 3 - \ln 10.$$

Also $\log_{10} 2 = \dfrac{\ln 2}{\ln 10} > \dfrac{3}{10}$, and since $\ln 10 > 0$,

$$\ln 10 < \frac{10}{3} \ln 2.$$

The first is an upper bound for $\ln 2$, and $\ln 2$ appears in the second with a positive coefficient, so we may substitute it:

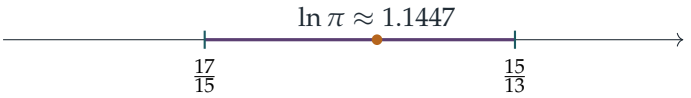
$$\ln 10 < \frac{10}{3}(3 - \ln 10) = 10 - \frac{10}{3} \ln 10.$$

Collecting the $\ln 10$ terms, $\frac{13}{3}\ln 10 < 10$, so $\ln 10 < \frac{30}{13}$. Therefore

$$\ln \pi < \tfrac{1}{2}\ln 10 < \tfrac{1}{2}\cdot\tfrac{30}{13} = \frac{15}{13},$$

as required.

Parts (ii) and (iii) together trap $\ln \pi$ in a narrow interval:



Checks

Numerically, $\ln \pi \approx 1.1447$, which does lie between $\frac{17}{15} \approx 1.1333$ and $\frac{15}{13} \approx 1.1538$. In part (i), $\log_\pi 10 \approx 2.0115$, only just above 2, which matches how close $\pi^2 \approx 9.87$ is to 10. The bound $\ln 10 < \frac{30}{13} \approx 2.3077$ is also sound against $\ln 10 \approx 2.3026$. It is worth confirming the given facts too: $\log_2(\pi/\mathrm{e}) \approx 0.209$, $\mathrm{e}^2 \approx 7.39$, $\mathrm{e}^3 \approx 20.09$ and $\log_{10} 2 \approx 0.3010$.

Common slips

- Writing $\dfrac{1}{\log_2 \pi}$ as $\log_2 \frac{1}{\pi}$. The reciprocal swaps the base and the argument; it does not invert the argument.
- In part (iii), using a lower bound for $\ln 2$ where an upper bound is needed. Chaining inequalities only works when each bound points the same way.
- Forgetting to say that $\ln 2$, $\ln 10$ and $\log_\pi$ behave well (positive, or increasing) before multiplying or taking logarithms across an inequality.

Beyond STEP

Proving numerical bounds from a handful of crude facts is a small taste of how analysts estimate constants: much of analysis consists of trapping an unknown quantity between two estimates you can control. Sharper bounds on $\ln 2$ or π come from series and integral inequalities, such as $\ln 2 = \int_1^2 \frac{1}{t}\,\mathrm{d}t$ estimated by rectangles. Abbott's *Understanding Analysis* is a friendly introduction to this way of thinking.

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